Dr. André Bauer

andre.bauer@chicagobooth.edu | https://bauer-research.net

Curriculum Vitae

Research Interest

My research is in performance engineering for AI and cloud systems, making the infrastructure behind modern AI efficient, predictable, and trustworthy. This work comprises 98 peer-reviewed publications (75 since my PhD in 2020) with 2,464 citations and an h-index of 25 (Google Scholar, October 2026).

Academic Career

University of Chicago Booth School of Business — *Chicago, United States*
Visiting Assistant Professor of Econometrics and Statistics — *since September 2026*

Illinois Institute of Technology — *Chicago, United States*
Assistant professor — *August 2024 — August 2026*

Argonne National Laboratory — *Lemont, United States*
Associated researcher — *January 2023 — August 2024*

University of Chicago — *Chicago, United States*
Postdoctoral scholar (Prof. Ian Foster) — *November 2022 — August 2024*

University of Würzburg — *Würzburg, Germany*
Postdoctoral scholar (Prof. Samuel Kounev) — *January 2021 — October 2022*

University of Würzburg — *Würzburg, Germany*
Research assistant — *October 2016 — December 2020*

School & Academic Education

University of Würzburg — *Würzburg, Germany*
Habilitation in Computer Science — *July 2022 — October 2026*

- **Thesis** "Enabling Efficient Scientific Computing: Intelligent Resource Management in the Age of Cloud and AI"
- **Mentors** Prof. Dr. Samuel Kounev, Prof. Dr. Ingo Scholtes, Prof. Dr. Rüdiger Pryss
- **Status** Thesis submitted December 2025, final faculty decision scheduled for October 22, 2026

University of Würzburg — *Würzburg, Germany*
Dr. rer. nat. (Computer Science) — *October 2016 — December 2020*

- **Dissertation** "Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases"
- **Advisor** Prof. Dr. Samuel Kounev
- **Grade** Summa cum laude (1.0) — *Defense December 22, 2020*

University of Würzburg — *Würzburg, Germany*
Master of Science in Computer Science — *October 2014 — September 2016*

Blekinge Institute of Technology — *Karlskrona, Sweden*
Semester abroad (Erasmus program) — *August 2015 — January 2016*

University of Würzburg — *Würzburg, Germany*
Bachelor of Science in Computer Science — *October 2012 — September 2014*

Awards & Scholarships

Best Industry Paper Award, ACM/SPEC International Conference on Performance Engineering (ICPE) — *Florence, Italy, May 2026*
For the paper "Evaluating Kubernetes Performance for GenAI Inference: From Automatic Speech Recognition to LLM Summarization".

Best Student Paper Award, IEEE International Conference on e-Science (eScience) — *Chicago, United States, September 2025*
For the paper "Efficient Privacy-Preserving Recommendation on Sparse Data using Fully Homomorphic Encryption" (co-author).

Best Teacher Award, Illinois Institute of Technology — *Chicago, United States, April 2025*
The department of computer science honors the best teacher every year.

GAIN Participation Grant, German Academic International Network — *Bonn, Germany*
The GAIN conference connects internationally mobile German scientists with representatives from the scientific community, politics, and industry. *May 2023*

Runner-up for Best Data Challenge Paper, ACM/SPEC International Conference on Performance Engineering (ICPE) — *Coimbra, Portugal*
For the paper "Efficient Data Processing: Assessing the Performance of Different Programming Languages". *April 2023*

2023 SPECtacular Award, Standard Performance Evaluation Corporation (SPEC) — *Gainesville, United States*
The SPECtacular Award is given to individuals who have significantly exceeded expectations in their contributions to SPEC. *February 2023*

Leadership Academy Fellowship, German Scholars Organization — *Berlin, Germany*
The Leadership Academy Fellowship is awarded to 25 outstanding researchers from all disciplines. *January 2023*

Best Industry Paper Award, ACM/SPEC International Conference on Performance Engineering (ICPE) — *Virtual conference*
For the paper "Why Is It Not Solved Yet? Challenges for Production-Ready Autoscaling". *April 2022*

SPEC Kaivalya Dixit Distinguished Dissertation Award, Standard Performance Evaluation Corporation (SPEC) — *Gainesville, United States*
The SPEC Kaivalya Dixit Distinguished Dissertation Award is an annual award recognizing one outstanding doctoral dissertation within the scope of the SPEC Research Group in terms of scientific originality, scientific significance, practical relevance, impact, and presentation. *April 2022*

Distinguished Dissertation Award, Faculty of Mathematics and Computer Science — *Würzburg, Germany*
The Distinguished Dissertation Award is an annual award for outstanding dissertations from the Faculty of Mathematics and Computer Science. *July 2021*

Dissertation Award, Unterfränkische Gedenkjahrstiftung für Wissenschaft — *Würzburg, Germany*
The Dissertation Award is an annual award for outstanding dissertations from all disciplines. *May 2021*

Summa cum laude, University of Würzburg — *Würzburg, Germany*
Grading of the doctoral thesis "Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases". *December 2020*

Winner of the FameLab State of Bavaria final, British Council — *Regensburg, Germany*
In this science communication competition, researchers must present their work in three minutes in an understandable and humorous way. *March 2020*

Sokrates/Erasmus Grant, University of Würzburg — *Würzburg, Germany*
Mobility grant for a semester abroad in Sweden. *August 2015 — January 2016*

Deutschlandstipendium, University of Würzburg — *Würzburg, Germany*
The Deutschlandstipendium supports talented and high-achieving students, taking into account not only first-class grades but also social commitment and special personal achievements. *October 2014 — September 2015*

Heinrich-Thein Award, Heinrich-Thein-Stiftung — *Haßfurt, Germany*
Award for talented students pursuing the "second-chance education". *November 2009*

Funding

NSF CRII Grant ($150,000), National Science Foundation — *August 2025*
PI for the project "Co-Simulation-Based Optimization of Container Orchestration Platforms".

Google Research Grant ($5,000), Google — *May 2025*
Acquisition of Google Cloud credits for the project "Assessing Microservice Applications Performance for Benchmarking".

Google Research Grant ($1,000), Google — *March 2023*
Acquisition of Google Cloud credits for the project "Performance Analysis and Prediction of Microservices in Cloud Environments".

DFG Research Grant ($347,235), German Research Foundation (DFG) — *January 2023*

Co-PI, with Prof. Samuel Kounev, for the project "MINERVA: Performance Prediction of Microservice Applications with Black-box Container Orchestration". The amount is my share of the grant.

Normalverfahren Grant 2021 ($3,234), Universitätsbund Würzburg — *April 2021*

Acquisition of the Normalverfahren Grant for the project "Prediction of Winter Mortality of Bee Colonies Using Machine Learning".

Teaching Activities

Analysis of Financial Time Series (BUSN 41921), University of Chicago Booth School of Business — *Chicago, United States* — *March 2027 — June 2027*

Lecturer of this MBA course (English), Spring Quarter 2027.

Advanced Business Statistics (BUSN 41001), University of Chicago Booth School of Business — *Chicago, United States* — *January 2027 — March 2027*

Lecturer of this MBA course (English), Winter Quarter 2027.

Time Series Analysis and Forecasting, Illinois Institute of Technology — *Chicago, United States* — *January 2026 — May 2026*

Lecturer of this self-created master's course (English).

Operating Systems, Illinois Institute of Technology — *Chicago, United States* — *August 2025 — December 2025*

Lecturer of this bachelor's and master's course (English).

Time Series Analysis and Forecasting, Illinois Institute of Technology — *Chicago, United States* — *January 2025 — May 2025*

Lecturer of this self-created master's course (English).

Distributed Systems, University of Chicago — *Chicago, United States* — *September 2023 — December 2023*

Co-Lecturer with Prof. Kyle Chard of this master's course (English).

Time Series Analysis and Forecasting, University of Würzburg — *Würzburg, Germany* — *April 2022 — September 2022*

Lecturer of this self-created master's course (English).

Advanced Programming, University of Würzburg — *Würzburg, Germany* — *October 2021 — March 2022*

Co-Lecturer with Prof. Samuel Kounev of this master's course (English).

Computer Science: Getting Started, University of Würzburg — *Würzburg, Germany* — *April 2021 — September 2021*

Co-Lecturer with Prof. Samuel Kounev of this bachelor's course (German).

Operating Systems, University of Würzburg — *Würzburg, Germany* — *October 2020 — March 2021*

Co-Lecturer with Dr. Nikolas Herbst of this bachelor's/master's course (German).

Software-Engineering Seminar, University of Würzburg — *Würzburg, Germany* — *October 2016 — March 2021*

Instructor of this bachelor's/master's course (German).

Operating Systems (Exercise), University of Würzburg — *Würzburg, Germany* — *October 2016 — March 2021*

Coordinator & instructor of this bachelor's/master's course (German).

Practical Course in Software, University of Würzburg — *Würzburg, Germany* — *October 2016 — September 2020*

Coordinator & instructor of this bachelor's course (German).

Student Supervision

PhD Students

- Laya Fakher, Illinois Institute of Technology. My PhD student until August 2026, now advised by Prof. Minxuan Zhou with me as co-advisor. — *since August 2025*
- Mark Leischner, Illinois Institute of Technology. My PhD student until August 2026, now advised by Prof. Nathaniel Hudson with me as co-advisor. — *since January 2025*
- SeongHo Parker, Illinois Institute of Technology. Left for industry. — *August 2025 — August 2026*
- Yannik Lubas, University of Würzburg. Co-supervised with Prof. Samuel Kounev in the DFG project MINERVA. — *since June 2024*
- Ivo Rohwer, University of Würzburg. Co-supervised with Prof. Samuel Kounev in the DFG project MINERVA. — *since June 2024*
- Three PhD students, University of Chicago. Co-supervised with Prof. Kyle Chard. — *November 2022 — August 2024*
- Four PhD students, University of Würzburg. Co-supervised with Prof. Samuel Kounev. — *January 2021 — October 2022*

Undergraduate and Master's Students

- Iqtedar Uddin and Jiya Sheetal Rathi, Illinois Institute of Technology, independent studies. *2025–2026*
- NSF BigDataX REU program, Illinois Institute of Technology. Co-PI, responsible for a cohort of ten undergraduates, two of them directly supervised. *Summer 2025*
- 19 bachelor's and 7 master's theses, University of Würzburg. *2016–2022*

Invited Talks

A Glimpse of the Argonne Leadership Computing Facility (ALCF) Inference Dataset *Chicago, US*
Parsl Fest, Parsl and Globus Compute Community Meeting *September 2026*

An Open Function-as-a-Service Dataset From the Edge to the Cloud *Chicago, US*
Parsl Fest *September 2024*

Academic Success: Navigating your Journey *Chicago, US*
University of Chicago *February 2024*

Smart Infrastructure Management: Navigating Efficiency in Scientific Computing *Chicago, US*
Illinois Institute of Technology *November 2023*

Why Auto-Scaling Still Matters *Chicago, US*
DePaul University *October 2023*

Towards an Intelligent Infrastructure for Data Science Clouds *Edmonton, Canada*
Chalmers University of Technology *May 2023*

Autonomous Data Analytic Clouds *Gothenburg, Sweden*
Chalmers University of Technology *February 2023*

A Short Overview of Auto-Scaling & Why It Is Not Yet Solved *Virtual meeting*
MongoDB *April 2022*

Measuring Energy-Efficiency in Digital Systems *Potsdam, Germany*
Hasso-Plattner-Institut *March 2022*

Academic Service

University Service

- Member of the faculty hiring committee for the Mumbai campus, Illinois Institute of Technology *May 2026 — August 2026*
- Chair of the CS Seminars Committee, Department of Computer Science, Illinois Institute of Technology *August 2025 — August 2026*
- Faculty co-advisor of the Illinois Institute of Technology chapter of Upsilon Pi Epsilon, the international honor society for the computing sciences *May 2025 — August 2026*
- Member of the PhD Recruitment and Experience Committee, Department of Computer Science, Illinois Institute of Technology *August 2024 — August 2026*

Organization Committee and Reviewing

- Workshop on Hot Topics in Cloud Computing Performance *2025*
- International Conference on Performance Engineering (Data Challenge) *2025*
- International Conference on e-Science (Posters) *2024*
- International Conference on Performance Engineering (Posters and Demos) *2024*
- International Workshop on Performance-Data Analytics and Data-Management *2023, 2022*

Program Committee and Reviewing

- International Conference on Utility and Cloud Computing *2026, 2025, 2024, 2023, 2022*
- International Conference on Time Series and Forecasting *2026, 2025, 2024, 2023, 2022*
- International Conference on Data Science, Technology and Applications *2026, 2025, 2024, 2023*
- Int'l Conference on Autonomic Computing and Self-Organizing Systems *2026, 2025, 2024*
- International Conference on e-Science *2026, 2025, 2024*
- International Conference on Performance Engineering *2027, 2026, 2025, 2024*
- International Workshop on Intelligent and Adaptive Edge-Cloud Operations and Services *2026, 2025*
- International Conference for High Performance Computing, Networking, Storage and Analysis *2026*
- International Conference on Software Engineering for Adaptive and Self-Managing Systems *2027, 2026*
- Workshop on Hot Topics in Cloud Computing Performance *2026*
- Symposium on Software Performance *2025, 2024, 2023, 2022*
- International Symposium on Cluster, Cloud and Internet Computing (Artifact Evaluation) *2025, 2024*
- International Conference on Service-Oriented System Engineering *2025*
- International Conference on Cloud Computing and Services Science *2024*
- International Symposium on Cluster, Cloud and Internet Computing *2023*

- Int'l Conference on Autonomic Computing and Self-Organizing Systems (Posters and Demos) *2023, 2022*
- International Conference on Performance Engineering (Artifact Evaluation) *2020*
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems *2019*

Associate Editor

- Frontiers in Digital Health, Health Informatics *since March 2026*

Editorial Board Member

- ICCK Journal of Systems Scalability *since October 2025*
- MDPI Data *since January 2025*

Financial Chair

- International Conference on e-Science *2025*

Publicity Chair

- International Conference on Performance Engineering *2020*
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems *2019*

Standard Performance Evaluation Corporation

- Founder and chair of the SPEC RG Predictive Data Analytics Working Group *since June 2021*
- Elected editor of the annual SPEC RG newsletter *April 2017 — May 2023*
- Elected release manager of the SPEC RG Cloud Working Group *October 2018 — May 2021*

Journal Reviews

- Journal of Systems & Software *2026, 2022, 2018*
- Journal of Parallel and Distributed Computing *2026, 2021, 2019*
- Journal of Grid Computing *2026, 2023*
- Journal of Systems Scalability *2026*
- Performance Evaluation *2026*
- Journal of Supercomputing *2025*
- Information Processing and Management *2025*
- Frontiers in Digital Health *2025*
- Empirical Software Engineering *2024, 2023, 2022*
- SoftwareX *2024*
- Transactions on Knowledge and Data Engineering *2023*
- Science of Computer Programming *2023*
- Transactions on Autonomous and Adaptive Systems *2023*
- Transactions on Cloud Computing *2021*
- Transactions on Services Computing *2020*
- Future Generation Computer Systems *2019*
- Transactions on Network and Service Management *2018*

Science Communication Competitions

1st South Side Science Festival Science Slam *Chicago, USA*
University of Chicago *September 2023*

6th GAIN Science Slam *Boston, USA*
German Academic International Network *August 2023*

21st Stuttgart Science Slam *Stuttgart, Germany*
LUUPS *October 2022*

FameLab State of Bavaria final *Regensburg, Germany*
British Council *March 2021*

6th Alumni-Science Slam *Würzburg, Germany*
University of Würzburg *November 2020*

FameLab German final *Bielefeld, Germany*
British Council *May 2020*

FameLab State of Bavaria final *Regensburg, Germany*
British Council *March 2020*

14th Munich Science Slam *Munich, Germany*
LUUPS *February 2020*

33rd Dortmund Science Slam *Dortmund, Germany*
LUUPS *November 2019*

Certificates

Mental Health First Aid USA National Council for Mental Wellbeing *March 2025*
Certificate of Leadership – Smart Skills National Postdoctoral Association *January 2024*
Certificate Leadership University of Würzburg *October 2022*
Certificate Academic Teaching – State of Bavaria University of Würzburg *November 2020*
Certificate Academic Teaching – University of Würzburg University of Würzburg *December 2019*

Voluntary Activities

Mentoring for high school students *Chicago, United States*
Illinois Science & Technology Coalition *August 2024 — August 2026*

Pen Pal Program *Chicago, United States*
Office of Diversity and Inclusion University of Chicago *February 2023 — August 2024*

Mentoring for students *Würzburg, Germany*
Alumni Network University of Würzburg *November 2019 — October 2022*

Supervision of scouts *Würzburg, Germany*
Schwarzer Haufen Würzburg (BdP e.V.) *July 2019 — February 2020*

Volunteer tutoring *Würzburg, Germany*
Nachbarschaftshilfe Grombühl *September 2013 — September 2014*

Student council *Schweinfurt, Germany*
Friedrich-Fischer Schule *September 2010 — August 2011*

Volunteer fire fighter *Horhausen, Germany*
FFW Horhausen e.V. *April 2003 — August 2009*

Languages

German Mother tongue
English Fluent